

BETA-CH-450 Air Cooled Chiller - Alarm and Fault Troubleshooting Guide

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Purpose

This guide covers the alarm and fault handling of the BETA-CH-450 air cooled screw chiller, 450 kW, serving the process cooling in the Line 1 machining cell.

The alarm code set of this controller is the F, H and E set listed under Alarm codes used by this controller. It is not the A-series set used on chiller controllers from other manufacturers. Confirm the code set from the controller display before using Alarm codes used by this controller, and never quote an alarm code from another unit's guide.

Technical data

Parameter	Value
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Nominal capacity, at rating condition	450 kW cooling at 45 degF leaving water, 95 degF ambient
Refrigerant	R-1234ze
Chilled water flow	82 m3/h
Chilled water, supply set point, operating value	46 degF
Chilled water, return, design	56 degF
Maximum ambient for full capacity	95 degF
Design condenser air flow	5.4 m/s across the coil
Glycol	inhibited glycol, 25 percent minimum by volume
Controller firmware	5.x series uses the F, H and E alarm codes
Freeze protection	trip when evaporator outlet water is 44 degF or lower
High head pressure trip	discharge at or above 217 psi

The nominal capacity of 450 kW is the manufacturer's rating condition: the duty the unit is certified to deliver at 45 degF leaving water and 95 degF ambient. The supply set point of 46 degF is the operating value the plant control runs to day to day. These two figures answer different questions and are not in conflict. The unit is rated one degree below its normal operating temperature, so at the 46 degF set point it delivers at least the rated 450 kW. A question about what the machine is capable of must be answered with the 45 degF rating condition; a question about what the plant is running to must be answered with the 46 degF operating set point.

The 95 degF ambient limit is a hard limit. Above it the unit cannot make its rated capacity and will eventually trip on high head pressure. A day when the site maximum ambient exceeded 95 degF must be recorded before the fault is attributed to the machine.

Alarm codes used by this controller

Code	Text on display	Meaning
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F-13	COMPRESSOR HP	Compressor tripped on high discharge pressure, 217 psi or above
F-21	OIL LEVEL LOW	Oil level switch open, or oil differential low
F-32	SUCTION LP	Suction pressure below the low limit
F-44	COMP STARTSW	Compressor failed to start within the start time
F-51	COMM LOSS	Controller lost communication with the plant module
H-04	E-HEAT STAGE	Electric heater stage fault or lockout
E-07	FREEZE STAT	Evaporator outlet water at or below 44 degF
E-12	FLOW SW OPEN	Evaporator flow switch did not prove
E-19	COND FAN	Condenser fan speed or feedback fault

A code that is not in this table is not documented here. Record the code, photograph the display, and raise a documentation request.

Section 4: the unit trips intermittently, repeated F-13

The most common report on this unit is "it trips at random". The code is F-13, high head pressure, and it has at least seven documented causes:

1. Ambient air above 95 degF. Check the site weather station reading for the trip time first. This is the most frequent cause and it is not a machine fault.
2. Condenser coil fouled with dust or process debris. Check the fin face.
3. Condenser fan failure or fan speed below set point. Check E-19 in the history.
4. Refrigerant overcharge, or a non-condensable gas in the system.
5. Glycol concentration below 25 percent, or the wrong glycol, causing poor heat transfer at the evaporator and high suction pressure.
6. Partially blocked evaporator plate heat exchanger from scale.
7. Safety valve lifting and not reseating, or a pressure transmitter out of calibration.

Discriminating steps, in order:

1. Read the site ambient temperature at the trip time from the history. If it was above 95 degF, the cause is external conditions; the machine is not diagnosed before the conditions are corrected.
2. Read the discharge pressure and the saturated condensing temperature from the trend history for the 15 minutes before the trip.
3. Check the condenser coil from the ground. Do not climb the unit.
4. Check the glycol concentration with a refractometer for the inhibitor package, not for ethylene glycol.
5. Compare the evaporator approach temperature. A large approach with a clean condenser indicates a refrigerant side or a glycol side problem, not a compressor problem.

Do not reset the unit and leave it. Three F-13 trips in one shift stops the unit until the cause is established.

Section 5: repeated E-07 freeze protection trips

Causes, in order:

1. Chilled water flow too low because a pump is on manual, a strainer is blocked, or a valve is closed.
2. Set point too low for the load. Check the supply set point against the design value in Technical data.
3. Thermistor drift. Compare the evaporator outlet water temperature on the control panel with a calibrated contact thermometer at the same point. A difference above 3 degF is a sensor fault, not a process fault.
4. Low glycol concentration or air in the water circuit.
5. Thermostat or capillary failure causing the valve to close late.

A freeze trip resets automatically once the water is above 48 degF. If it repeats, the automatic reset is not an acceptance criterion; establish the cause first.

Section 6: E-12 flow switch open

Causes: pump on manual, strainer blocked, flow switch not proved, valve closed, or a pump in the parallel pair that has failed.

Check the pump status and the strainer differential first. A flow switch failure is confirmed only after the flow path has been proven clear.

Section 7: F-21 oil level low

Causes: oil level switch fault, genuine oil loss from a leak, oil carryover to the evaporator, or an oil drain valve left open.

Confirm the oil level in the compressor sight glass first. If oil is present and the switch is open, the switch is faulty; if oil is low, find the leak. An oil drain valve left open has caused two F-21 calls on this unit.

Section 8: H-04 electric heater stage fault

The unit uses an electric heater stage in the evaporator for low load control and freeze protection. Causes: open heater element, failed contactor, failed high limit thermostat, or a control fault.

Check the high limit thermostat first: a stuck closed thermostat presents as a heater stage fault with a healthy element. The heater element resistance values are in the vendor manual and are not in this document collection.

Escalation

Escalate to Maintenance Engineering and the controls engineer when: three trips of the same code occur in one shift; a trip cannot be explained by the ambient conditions or the set points; the controller firmware does not match Technical data; or the required figure is not in the controlled document set. The

work order carries documentation status "insufficient information" in those cases.

Documentation gaps

Not contained in this guide:

- The alarm code map for controllers with firmware 4.x and earlier, which used a different code set. The map is Appendix 3 of the vendor manual, which is not in this document collection.
- The refrigerant charge quantity for this unit, which is on the equipment nameplate.
- The glycol inhibitor package specification and the interval for adding inhibitor.
- The pressure transmitter calibration procedure.
- The heater element resistance values.

Appendix: Extended alarm and fault matrix

This matrix extends the alarm codes and symptoms in the main sections with the verification steps and the most likely cause for each.

Code	Verification step	Most likely cause	First action
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F-13	Read ambient and discharge pressure	Ambient above 95 degF	Record conditions, do not reset
F-13	Check the condenser coil	Coil fouled	Clean the coil
F-21	Read the oil level at the sight glass	Low oil level	Top up and find the leak
F-32	Check the suction pressure	Low charge	Leak test and recharge
F-44	Check the compressor start sequence	Start fault	Investigate the contactor
F-51	Check the communication cable	Module fault	Reset or replace the module
H-04	Check the heater stage and high limit	Stuck high limit	Replace the thermostat
E-07	Compare the thermistor to a contact thermometer	Thermistor drift	Replace the thermistor
E-12	Check the flow switch and strainer	Strainer blocked	Clean the strainer
E-19	Check the fan feedback	Fan or feedback fault	Repair the fan circuit

A code that is not in the table is recorded with a photograph and reported as an undocumented code. The controller firmware version is recorded with the code because the code set changes between versions.

Appendix: Seasonal and condition-based checks

Chiller faults cluster around weather and load. This appendix lists the checks brought forward when the season changes or when the load changes.

Before the warm season:

- Clean the condenser coil and confirm the airflow is unobstructed.
- Test the condenser fans and confirm the speed feedback.
- Confirm the refrigerant charge by the subcooling and superheat method.
- Confirm the glycol concentration for the expected minimum temperature.
- Test the high-pressure trip and confirm the set point.
- Confirm the ambient sensor reads correctly against a reference.

Before the cold season:

Confirm the freeze protection set point and test the trip.
Confirm the glycol concentration for the lower ambient.
Confirm the heater stage, if fitted, operates and the high limit is set.
Confirm the water circuit is protected against freezing when idle.

After a load change:

Re-check the flow against the design value.
Re-check the approach temperatures against the baseline.
Re-check the compressor current against the nameplate.
Re-check the controller set points against the design values.

Each check is recorded with the date and the measured value. A check that is not recorded is treated as not done.

Appendix: Refrigerant, instrumentation and electrical reference

Item Value
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Refrigerant R-134a, charge 31 lb
Refrigerant top-up by weight only, never by pressure
Low pressure switch 29 psi cut-out, 51 psi cut-in
High pressure switch 348 psi cut-out, manual reset
High head pressure alarm F-13
Running current, compressor 42 A per phase
Supply 460 V, 3 phase, 60 Hz
Ambient limit 95 degF
Leaving water set point 46 degF
Freeze protection 48 degF reset
Glycol minimum 25 percent by volume

Refrigerant is added by weight from a calibrated cylinder and the quantity is recorded in the refrigeration log. Adding refrigerant until the suction pressure looks right is prohibited because it hides a leak and overcharges the circuit. A circuit that needs a top-up twice in a season is leak tested before it is returned to service.

Instrumentation used for fault finding must be within its calibration date.
Pressure gauges are read at the service port with the schrader core depressed for no longer than necessary. Clamp meters are placed on a single conductor; placing the clamp around a cable carrying both supply and return reads zero and is not a valid measurement.

Revision history

Revision Effective date Summary
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Rev 1 2024-09-09 First controlled issue, compiled from Line 1 fault history 2022 to 2024.